

AI LIVE OVERSIGHT

AI Live Oversight — all AI on, with a human able to watch, stop, and correct

All AI functionality runs autonomously from the get-go. This system gives a human real-time visibility and control without slowing the AI down, and it routes every change through the site's own users first.

The change lifecycle (individual → statistical → global)

1. **The AI proposes a change.** For any non-sensitive setting, it does NOT apply globally. It opens a change-gating experiment (`createExperimentForProposal`).
2. **Every user is asked yes/no.** The change is put to real members as a simple “does this work for you?” prompt (`ExperimentVotePrompt`, mounted globally; one vote per user via `submitExperimentFeedback`).
3. **The users' yeses and noes decide.** `evaluateExperiments` promotes a change only when it clears a high statistical bar: a minimum number of votes (`CHANGE_GLOBAL_MIN_SAMPLE`, default 20), a minimum yes-rate (`CHANGE_GLOBAL_MIN_APPROVAL`, default 0.70), AND a Wilson 95% lower-bound ≥ 0.5 so a small lucky sample can't win.
4. **AI conducts its own review and promotes — by default.** With the optional human gate OFF (`AI_GLOBAL_HUMAN_GATE`, default off), the AI sanity-checks the user-approved change (`aiReviewPromotion` — defaults to approve since users already said yes, but can HOLD something risky/unreliable) and pushes it site-wide. Each promotion is logged to the live feed.
5. **Optional human gate.** Flip `AI_GLOBAL_HUMAN_GATE` on and promotions instead wait for a once-per-24h, one-hour peak-time review window (`PEAK_REVIEW_HOUR_UTC`, `PEAK_REVIEW_WINDOW_HOURS`), where an admin promotes/rejects eligible changes (`aiGlobalReview` / `aiGlobalDecide`, window-gated).

Money/compliance settings are NEVER auto-tuned — the optimizer's `COMPLIANCE_DENYLIST` excludes them from this entire path.

Global controls (backend/sdk/ai-control.ts)

- **STOP button** — the `ai_paused` global kill switch. `aiPaused()` is checked by the autonomous loops (`runOptimizationPass`, `selfLearningCycle`, `KYC autopublish`), so engaging it instantly halts AI-driven changes. `setAiPaused()` flips it.
- **Real-time feed** — `logAiAction()` writes every AI action to `AIActivityLog`; `recentAiActivity()` reads it back. The optimizer and KYC AI are instrumented; the helper is available for any other engine.
- **Corrections that teach** — `recordCorrection()` records a human fix as an `AICorrection` plus a heavily-weighted `OptimizationSignal` and an `AgentLearningMemory` lesson, so the AI grounds on the fix on its next pass. Corrections work even while the AI is paused.

Functions & UI

- aiControlStatus (feed + paused state), aiControlPause (stop/resume), aiCorrectionSubmit (apply a fix to a safe setting + record the lesson), aiGlobalReview / aiGlobalDecide (optional daily gate).
- **AI Live Oversight** admin page: live feed (polls every 5s), the STOP/Resume button, per-action “Correct” flow, and the (optional) daily global-review panel.

Flags & settings

Flags: ai_paused (kill switch, default OFF), kyc_survey_ai_autopublish (AI applies live, default ON). Settings: AI_GLOBAL_HUMAN_GATE (default OFF), CHANGE_GLOBAL_MIN_APPROVAL (0.70), CHANGE_GLOBAL_MIN_SAMPLE (20), PEAK_REVIEW_HOUR UTC (18), PEAK_REVIEW_WINDOW_HOURS (1), OPTIMIZER_REQUIRE_EXPERIMENT (on — the per-user consent path).

Entities

AIActivityLog, AICorrection (both added to schema.sql + entities.json).